A Feasibility-Aware HUBO/QUBO Benchmark for UAV Obstacle-Avoidance with Visibility Cost

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Abstract

Quantum-assisted UAV path planning has recently been studied using QAOA-style obstacle-avoidance formulations, but a low-energy binary sample is not necessarily a valid UAV trajectory. This paper presents a feasibility-aware benchmark for a discrete 8×8, $L = 20$ UAV grid-planning problem with obstacle avoidance, a line-of-sight visibility cost, and a higher-order temporal buffer-risk term. The horizon L denotes discrete planning layers, not physical seconds. The model uses binary variables $x_{t,v}$ indicating whether the UAV occupies cell v at time layer t and forms a HUBO objective with one-hot, start, motion, obstacle, visibility, terminal-goal, and sustained near-obstacle proximity terms. The instance has 1,280 original binary variables and 118,344 polynomial terms, including 4,392 cubic monomials. We compare A*, RRT-best, a QUAV-style QAOA candidate-path selector, native trajectory-level simulated annealing (SA), the classical `neal` BQM sampler after HUBO-to-QUBO reduction, and an exact CP-SAT baseline that minimizes the native HUBO soft objective over hard-feasible paths. The CP-SAT solver returns OPTIMAL and certifies that the best observed SA path is globally optimal for the enforced feasible-path objective, with implementation HUBO energy -20752.43, soft cost 247.57, path length 14, six turns, and 85% visibility. The results do not claim quantum speedup; they show that feasibility-aware decoding, exact baselines, and the effects of quadratization are essential for interpreting quantum-compatible UAV planning results.

Keywords: HUBO, QUBO, feasibility-aware benchmarking, UAV path planning, obstacle avoidance, visibility-aware planning, CP-SAT, QAOA, quantum-compatible optimization.